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Agents and Pervasive Computing Group

Browsing the World in the Pervasive Computing Continuum:

Challenges and Directions for Middleware Research

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Motivations

- Computer-based systems and sensors will be soon embedded in everywhere
  - all our everyday objects
  - all our everyday environments
- Current deployments of pervasive services and sensor networks focus on special purpose systems, e.g.,
  - E.g., environmental monitoring and healthcare
- General purpose approaches are likely to emerge soon
  - Shared infrastructures of sensors, tags, cameras, Web 2.0 data
  - Collecting general purpose data about physical and social world
  - Defining a sort of distributed "***computational world model***"
  - For general-purpose exploitation by "***browsing the world***" services



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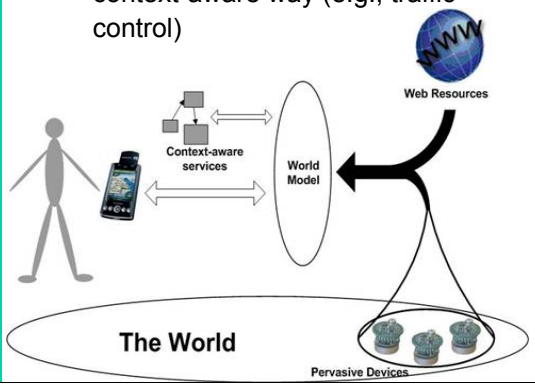
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"Browsing the World…"

- The computational world model as an infrastructure for services to:
  - *Interact* with the physical world (e.g., personalized real-time traffic maps)
  - *Act* in the physical world (e.g., actuating green waves in the traffic)
  - *Coordinate* with each other in a context-aware way (e.g., traffic control)





PLEASE NOTE: Users and services are themselves part of the world, and thus must be part of the computational world model



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"…in the Pervasive Computing Continuum"

- Distributed computer-based sensors and tags
  - Potentially very densely deployed in the environment
  - A "spatial continuum" of computational, sensing, and storing capabilities all around us
  - Generating a "temporal continuum" of information (streams)
  - Or, in any case, a variable density of sensors around, somewhere very dense, somewhere not
- Cameras, Social tagging, Geospatial information
  - Even if not "continuously" distributed in the environment
  - Can generate information about the continuum
  - Can store spatially situated information
  - And mobile actuators can enforce actions in the continuum
- Overall….
  - Future generation "Pervasive + Web" infrastructures
  - Can be perceived as a sort of "spatio-temporal" continuum of computational power, memory storage, and sensing capabilities



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Middleware Challenges

- How can we support the execution of services in the pervasive computing continuum?
  - This is definitely a "middleware challenge"
- Key (strictly inter-related) challenges that we will analyse:
  - Supporting adaptive service behaviour to tackle the dynamics of the scenario
  - Supporting context-awareness in the pervasive computing continuum
  - Supporting self-organized service composition
  - Supporting security and privacy



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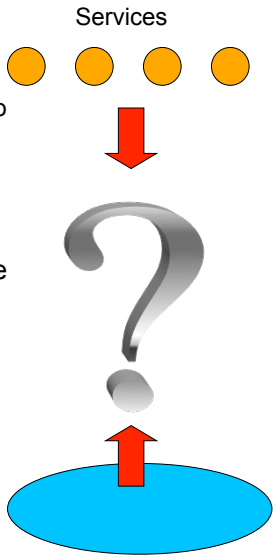


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Why Self-adaptation?

- The complexity, openness, and dynamics of the scenario makes it impossible for humans to stay in the control loop
  - Services must tune their behavior and self-adapt to the current world situation
  - Humans can intervene only on limited portions of the scenario
- But…..*is the challenge really with services*?
- Browsing the world services lead to a perspective of situated interactions
  - Services lives in and access the common "world model" environment
  - Act by sensing it and affect it by acting
  - The environment has its own dynamics and processes
- To enforce adaptive services, we could think at embedding self-adaptation in the computational world model, rather than on services
  - **Engineering the environment** (i.e., the computational world model) rather than services





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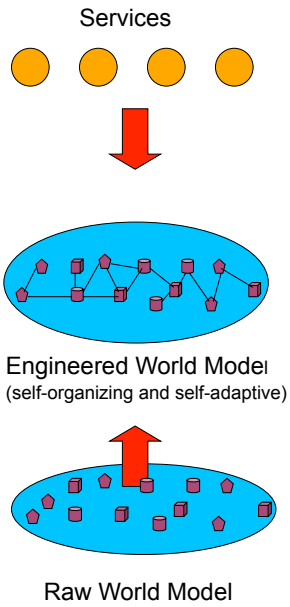


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Engineering the World Model

- Extreme heterogeneity of data
  - Sensors, cameras, Web 2.0, etc.
  - Variable density (potentially continuum)
- Massive amounts of data produced
  - No way to collect all data at a place
  - Need to aggregate, prune, evaporate, analyse data where it is produced
- Inherent dynamism and decentralization
  - Mobility & ephemerality: devices/data come and go at any time
  - No way to control each device due to decentralization
  - No user/services can be ensured the availability of specific data
- Need to adaptively provide services with specific views of the world model, to ensure they can go on at the best with their activitiies
  - Some sort of middleware should provide that….





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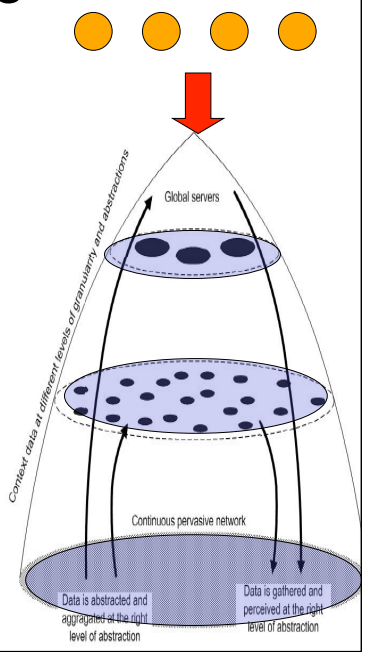


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Context-awareness in the Pervasive Computing Continuum

- Data virtually generated in both
  - A sensor network continuum
  - A level of Web 2.0 services
  - At any level in between (e.g., cameras, loca servers, etc.)
- How can services can get effective access to all needed information in such a messy environment?
- Data should flow up the pyramid
  - In aggregated forms, via proper self-organizing algorithms for data aggregation
  - To limit the amount of data (potentially infinite) to be managed by
- Data should flow down the pyramid
  - To self-aggregate data from low-level sensors with data from higher-level ones
  - To let agents exploit all available data in a uniform way and locally





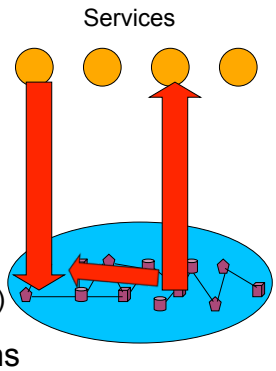
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Self-org Service Composition



- Self-organization via direct interactions
  - Components (services/agents) interact directly with each other to reach common goals in a self-organizing way
  - Not suitable to open and dynamic systems (who are the others?)
  - Not suitable to situated activities (where most issues relates to what's happening in the world, rather than what others are doing)
- Self-organization via mediated interactions
  - Self-organization takes place via sensing/affecting a common environment (stigmergic interactions, as in ant colonies)
  - The environment has its own properties and processes which rules the behavior of the colony (diffusion and evaporation of pheromones)
  - Components do not need to know each other (ants are blind)
  - Self-organization is by definition situated
- So, once again, we get back to **engineering the environment** rather than services



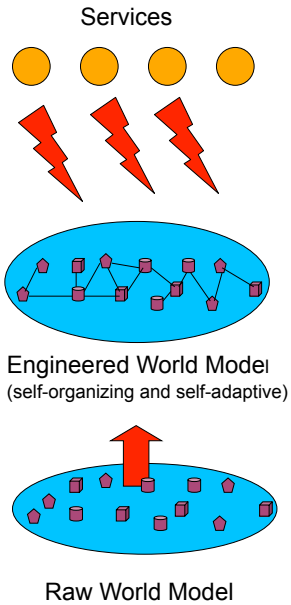
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Re-thinking Security and Privacy



- The pervasive computing continuum will be a shared infrastructure
  - Any user/service can contribute data to it
  - Any user/service can access it to perform its specific activities
- This clearly raises security and privacy issues
  - Who control the information? How can we ensure that information is not destroyed, manipulated, etc.
  - Who can access to what information? How can I ensure some sensible data is not made public?
- Again, the middleware should take care of these issues in its provisioning to services of the engineering computational world model



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Some Promising Directions

(from my own viewpoint, of course)

1. Algorithms for self-organized data aggregation in the pervasive computing continuum
2. Uniform general-purpose data models for situated services
3. General-purpose situated service models
4. Rethinking security in self-adaptive terms



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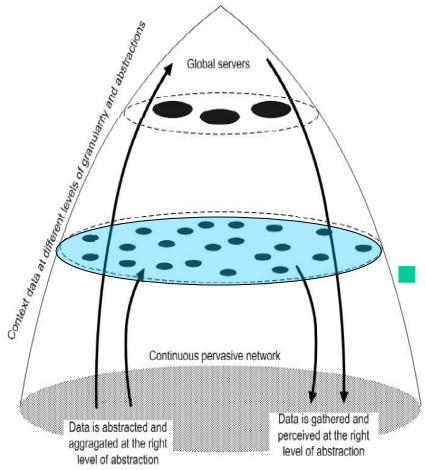


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Self-organizing Spatial Regions

(Bicocchi, Mamei, Zambonelli, IEEE SASO 2007)



- We need to aggregate data from the continuum
  - Compact representation, manageable by services
  - Without losing relevant information
  - And indeed provide more information of "what's happening"
- Key idea
  - Identify discrete aggregation regions
  - Enable "per region" views of specific characteristics of the environment
  - As if there were sort of virtual macro sensors



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Identifying Aggregation Regions

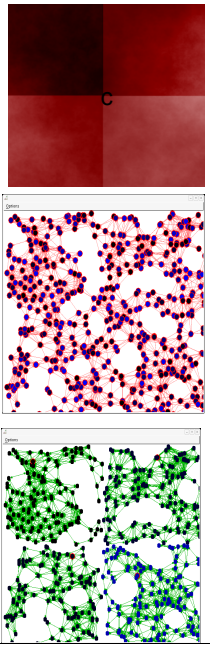
- Our "Region Aggregation Noise" (RAN) approach, considers the following:
  - A distributed algorithm is continuously running in the network as a sort of *"background noise"*
  - with the goal of partitioning the sensor network into regions characterized by similar patterns (as an overlay of virtual links)
  - Abstracting from the specific density/structure of the sensor network

```
Do_forerever:
      Wait(t);
      neigh[] = Select_neighbor(num_neigh);
      Foreach(neigh[])
            Data = Exchange_data();
            Update_link(data);
Done
```

```
Update_link:
if D(v(si),  v(sj)) < T {
      l(si,sj) = min(l(si,sj) + delta, 1)
} else {
      l(si,sj) = max(l(si,sj) - delta, 0)
}
```

Simulated Region Formation

- Simulated sensors are embedded in an environment characterized by different patterns of sensed data
- At first they are not logically connected with each other
- Gradually they recognize regions
- Then a partitioning emerges based on the environmental patterns



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Per-region Data Aggregation

- Once regions are formed,
  - *Aggregation of sensed data* can occur on a per-region basis
  - Simply by injecting a self-organizing (gossip-based) aggregation function on the network
  - That exploit the existing aggregation noise without incurring in additional communication costs
  - Users/services can, on need, be provided with such aggregated data representing some "macro" property of a region at very limited costs
- Example: Minimum Value in a Region

```
Do_forerever:
  Wait(t);
  neigh[] = Select_neighbor(num_neigh);
  Foreach(neigh[])
          Data = Exchange_data();
          Update_link(data);
          if(connected) Local_aggregation();
Done
```

```
Local_aggregation:
if(localMini>localMinj)
          localMini=localMinj
```



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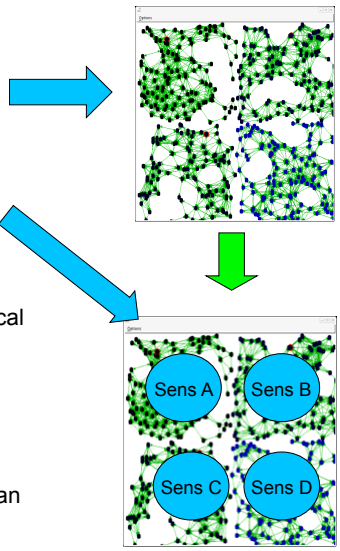


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Virtual Macro Sensors

- Eventually…
  - Aggregation of data can take place on a per region basis: all sensors in a region act as a single entity
- Which implies
  - Services can start perceiving the sensor network as composed of a finite number of macro sensors
  - Abstracting from the actual structure and density (potentially continuum) of the physical network
  - Rather focussing on the real environmental characteristics
- And also:
  - And the network of virtual macro sensors can automatically adapt to changes in the environment
  - Several virtual partitionings of regions can co-exists, to serve the needs of diverse services





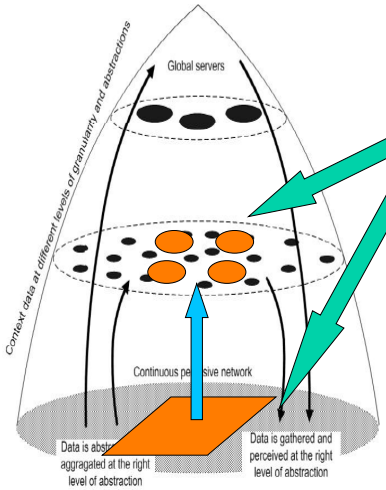
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But This is not Enough…



- The presented idea
  - Act on the sensors' continuum
  - To define a manageable virtual macro sensors level
- But we also need to account for
  - Integrating diverse and heterogeneous information
  - Coming from a variety of devices
- Enabling to generalize data aggregation to other dimensions
  - semantics, temporal, application specific
  - other than spatial
- To this end, a ***general data model*** is needed
  - Let's also talk about "knowledge"



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Data/Knowledge Models

- How can we provide a uniform representation of data (i.e., facts about the world, that is knowledge about it)
  - Coming from diverse devices (sensors, tags, cameras, and why not Web 2.0 fragments such as geo-tags)
  - Expressing different levels of observations (e.g., a single light sensor vs. a camera)
  - Expressing in a uniform way different redundant perspectives on the same fact (e.g., a WiFi vs. a GPS localization information)
  - Easy to be accesses and manipulated, aggregated (by both services and by within the world model)
- Key guidelines
  - Keep it simple
  - Keep it intuitive
  - Keep it computable
- How do we usually characterize facts about the world?
  - A possible idea: the W4 model



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The W4 Model

(Castelli, Mamei, Zambonelli, J. Inf. Soft. Tech., Jan. 2008)

- Someone or something (*Who*) does some activity (*What*) in a certain place (*Where*) at a specific time (*When*)

- **Who** is the subject. It is represented by a string with an associated namespace that defines the "kind" of entity that is represented.
  - "person:Gabriella", "tag:tag#567", "sensor-region:21"
- **What** is the activity performed. It is represented as a string containing a predicate-complement statement.
  - "read:book", "work:pervasive computing group", "read:temperature=23".
- **Where** is the location to which the context relates.
  - (longitude, latitude),
  - "campus", "here"
- **When** is the time duration to which the context relates
  - 2006/07/19:09.00am - 2006/07/19:10.00am
  - "now", "today", "yesterday", "before"



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W4 Knowledge: Atoms & Queries

- Gabriella is walking in the campus' park. An agent running on her PDA can periodically create an atom describing her situation.
  - **Who**: user:Gabriella
  - **What**: works:pervasive group
  - **Where**: lonY, latX
  - **When**: now

- Gabriella is walking in the campus, and wants to know if some colleague is near. She will ask (read operation):
  - **Who:** user:*
  - **What:** works:pervasive group
  - **Where**: circle,center(lonY,latX),radius: 500m
  - **When:** now

- Key features
  - Partial knowledge (not fully filled atoms)
  - Context-aware queries ("here", "now")
  - Possibility of generating atoms merging different sources
    - E.g., and RFID atom generated by merging tagID (who), DBMS data (what), GPS data (where), PDA data (when)
  - Device independence



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Self-organized Knowledge Networks

- Knowledge atoms (W4, or whatever) can be related to each other
  - To represent relations between pieces of data and enable navigating related facts of the world
    - Spatial, temporal, or semantic relations
  - To support a better navigation of services in knowledge
  - And more “cognitive” forms of self-organization in services
- Can we exploit self-organization approaches?
  - Self-aggregation of data via semantic/temporal extension of the spatial region approach (cluster and link related data to define multiple and multilevel knowledge views)
  - Have data diffuse (data distribution) and evaporate (data obsolescence) the same as pheromone does in ant-based systems
  - Create knowledge structures that services can perceive as sorts of virtual force fields (context-aware data replication)
  - Matching data patterns and chemical-like reactions (creation of new knowledge)
- A lot of issues to be investigated
  - But ***what service model*** can take advantage of it?



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What is a Situated Service for Browsing the World?

- Browsing the world services
  - Location-dependent queries, Social interactions, Real-time understanding of situations, etc.
- Autonomic communication services
  - Distributed cooperation, Ad-hoc communications, etc.
- Given the availability of W4 spaces and data (we assume that there are a multiplicitly of accessible “W4 space” where to store locally produced W4 tuples),
  - we have to code specific software components that can somewhat “query” the W4 tuple spaces for
  - Achieving context-awareness and adapt to the current situation
  - Navigating the world model and extract high-level information about situation occurring in the environment
  - Coordinate with each other in a self-organizing way
- So it is a **software agent**



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Key Features of Services/Agents

- We require services the capability of
  - querying the local world model for extracting info (also as subscriptions to events)
  - injecting new data/knowledge atoms and/or new aggregation algorithms in the space

```
KnowledgeAtom[ ] read(KnowledgeAtom template);
KnowledgeAtom[ ] subscribe(KnowledgeAtom template);
void inject(KnowledgeAtom a);
```

- Also consider that:
  - Services/agents themselves (their characteristics and actions) will be represented by some sorts of atoms in the space
  - In the end, in a “smart objects world”, objects can generate data and that, by accessing data and by hosting computations, can implemented coordinated services
- After all, this may appear to simply reduce to a “shared blackboard” or “tuple space” interaction model
  - Well, actually, it is more like and “ant-based” self-organization model



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Browsing the World: Services vs. Ants

- Services produce and read W4 tuples
  - The same as ants release and sense pheromones
- Services indirectly interact via the W4 distributed environment
  - Stigmergy!
- The environment describe something about the environment
  - "There is another agent near here"…the same as pheromones do
- The environment is active
  - Aggregation and pruning of W4 atoms → similar to diffusion and evaporation of pheromones
- We still have to fully unfold these issues, and define a simple and usable agent model…
  - Yet, we have a prototype infrastructure…



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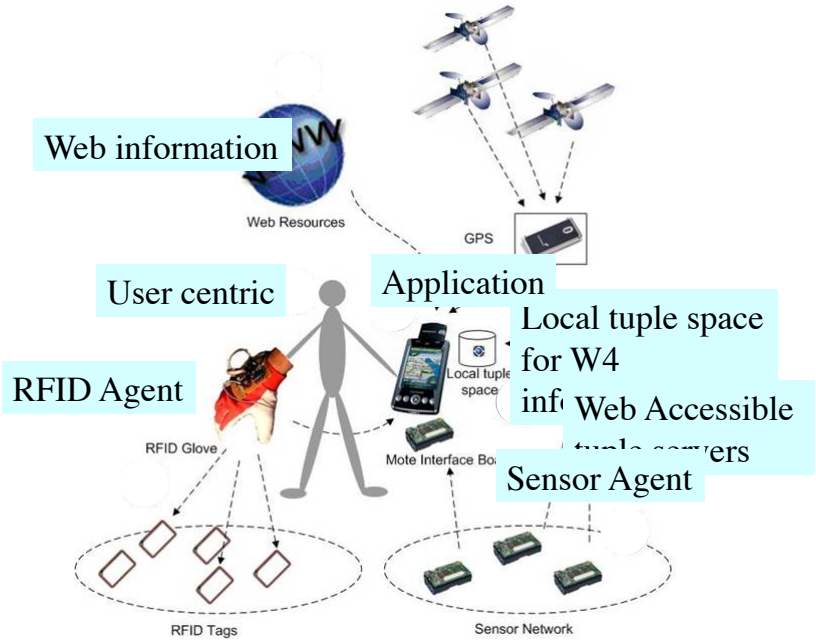


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Browsing the World Architecture

(Castelli, Rosi, Mamei, Zambonelli, IEEE PerCom 2007)





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Implementation

- The whole system has been realized using the Java language.
- **Web (global accessible) Tuple Spaces** has been implemented through a Postgres database with spatial extensions.
- The **Local Tuple Space** is simply implemented by a Java Vector.
- The RFID reader and the Mote Sensors are accessed via JNI and TCP/IP.
- The **"W4 Query Engine"** interrogates the web accessible tuple space through Sql and its postgis spatial extensions. Local tuple space instead makes use of String parsin and java algorithm.
- User interface is provided by:
  - Google Earth (for laptops) and Google Maps accessed via the Minimo browser (for PDAs).
  - Google KML Language
  - Jsp e JavaScript



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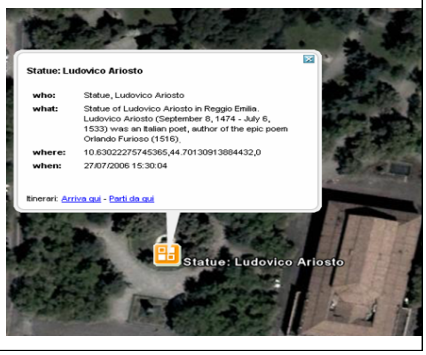
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Application: The Journey Map

- A tourist wants to automatically build and maintain a diary of his journey:
  - track of all the user movements
  - access available tourist information stored in RFID tags attached to monuments and art-pieces

- **Who:** rfid:*
- **What:** *
- **Where:** *
- **When:** now







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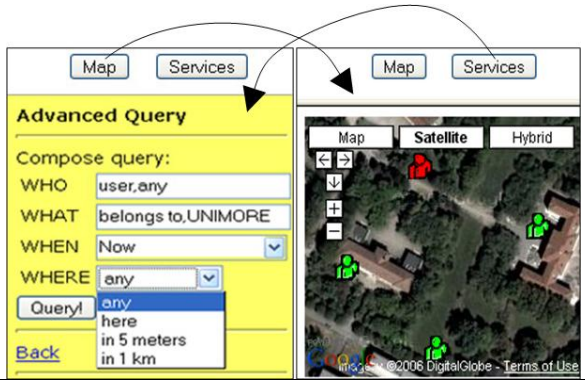


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Application: Real Time Maps

- Friends/members of a rescue time can share their actual GPS locations (W4 knowledge atoms)
  - They can be used to see each other
  - And to coordinate their movements with each other and with the environment





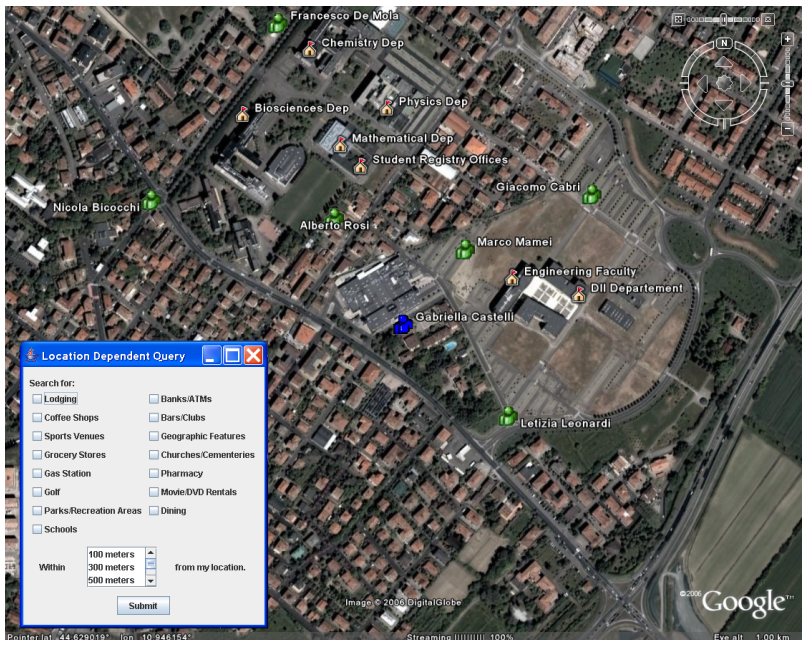
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Application: Real Time Maps





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Security Issues

(please note I'm not an expert on security!!!)

- Key problem:
  - Data affect services behavior at both individual and collective level
  - How to protect data (and the objects/devices from which such data come) from attacks?
- Provocative question: *is this really an issue?*
  - If users/developers have no direct control over the whole system, but only over limited portions of it, how can hackers have?
  - If data is really overwhelming and to most extent intertwined and redundant, how can hackers significantly affect information?
- Examples:
  - Localization: GPS, WiFi, Bluetooth, Mobile Telephony, RFID tags
  - Who can really affect all of these to generate believable localization information?
  - (think also at citation indexes: would you be able to affect both SCOPUS, CiteSeer, DBLP, GoogleScholar, to make people really believe you are a highly cited author?)



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Security vs. Disturbances

- We should try thinking at security in totally different terms
  - Attacks can be perceived as a background noise, a disturbance on the perception of the "world model"
  - Self-organizing services should exhibit dynamics capable of tolerating disturbances
    - As already said, they should be able to deal with partial information, as well as with redundant information
    - They should be able to tolerate inconsistencies
  - Calling for self-organized aggregation algorithms for data redundancy and data consistency against noise
- What about privacy?
  - Do not know, I am tempted to think it is mostly hopeless to strive for it



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Conclusions

- Future pervasive computing scenarios invites considering
  - A continuum of sensors making available a sort of world model
  - Self-organizing and adaptive services for browsing the world
  - Relying on the self-organizing capabilities of a computational world data model, intended as a virtual structure made available by some middleware infrastructure
- Fulfilling the vision is very challenging, calling for
  - General self-organizing algorithms for data/knowledge aggregation
  - Proper general models for representing, accessing, and integrating heterogeneous contextual data
  - Suitable service models exploiting the above for the provisioning of autonomic services
  - Novel, non traditional. perspectives on handling security
- And many other issues I have not identified….



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Get Involved…

- PANORAMA: FP7-FET Coordination Action on "Pervasive Adaptation"
  - Pervasive Computing + Software Adaptaption
  - Working Groups
  - Summer School (Rimini, Sept. 6-13 2008)
- IEEE SASO 2008 Conference: "Self-adaptive and Self-organizing Systems"
  - Tutorials, Workshops, Technical Papers (deadline around May)
  - Workshop on Pervasive Adaptation